

Labour of Love: Volunteer Perceptions on Building Relatedness in Online Volunteering Communities

Khushnood Z. Naqshbandi

Affective Interactions Lab, School of Architecture, Design and Planning, The University of Sydney

Ajit G. Pillai

Affective Interactions Lab, School of Architecture, Design and Planning, The University of Sydney

Silas Taylor

Office of Medical Education, University of New South Wales

Naseem Ahmadpour

Affective Interactions Lab, School of Architecture, Design and Planning, The University of Sydney

ABSTRACT

Volunteerism in the digital age offers many new possibilities and avenues for public participation. In this paper, we discuss volunteering as a form of work and how certain experiential aspects of HCI systems supporting the volunteers' unpaid labour are instrumental in volunteer wellbeing. The sense of feeling close to others and experiencing relatedness is an important factor that can predict engagement and wellbeing of volunteers. Relatedness can be achieved in many ways, for instance, through expressing and receiving gratitude. Through four co-design workshops with $n=9$ participants, we identified seven perceptions of volunteers regarding their relatedness experiences. This was achieved via a case study of an online teleconferencing platform where volunteers help train and assess medical students for their medical communication skills. Findings are further discussed to inform future design to support adequate level of formality and emotional labour in online volunteering communities.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI.

KEYWORDS

Volunteering, Work, Relatedness, Gratitude

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1 INTRODUCTION

The field of Human-Computer Interaction (HCI) is replete with research on supporting work, performance, productivity and worker's

wellbeing. Recent research in HCI emphasizes the need for 'worker-centric design', an approach to design for supporting workers' interests beyond the workplace context in order to build a better future of work [1]. Examples of work in HCI range from platforms used for professional work in corporations to informal work in paid gigs supported by apps such as Uber and online platforms like Amazon Mechanical Turk. These are all instances of work in the paid economy as opposed to the unpaid economy represented by volunteers – the amateurs, enthusiasts, and do-gooders. Volunteers are driven by things other than the material rewards of the work. In G.K. Chesterson's words, "Such a man must love the toil of work more than any other man can love the rewards of it" [2].

Volunteers engage in a lot of important work. An example of volunteer work is Wikipedia, the free online encyclopedia, which has very famously usurped the dominance of professionally created encyclopedias such as Encarta and Britannica. Not only do volunteers contribute billions of dollars' worth of labour to the global economy, but such work is also often very important in enabling, maintaining and sustaining essential processes that run the economy.

HCI research provides evidence on the drivers of volunteering and what improves volunteer productivity and wellbeing. For instance, volunteer wellbeing is linked to their experiences of relatedness [3]. Relatedness – the sense of being connected to others – has been linked to improved happiness and wellbeing at workplace [4][5]. Along similar lines, Sirota et al identified camaraderie at workplace in their extensive research involving over four million employees globally as one of the three main ways of maintaining a happy worker [6]. One way of fostering relatedness among volunteers is via expressions of gratitude [3][7]. Cultivating gratitude in workplace settings has been associated with improved relatedness and other subjective wellbeing markers [8][9]. In fact, a study on professionals at a nursing school showed that cultivating an 'attitude of gratitude' improved employee retention and job satisfaction by 17.9% [10].

Our previous research on online volunteerism revealed the importance of relatedness for improving volunteer's motivation and wellbeing, with gratitude being the mediating factor [3]. In this paper, we present our work-in-progress on how these abstract concepts are expressed in an online volunteering platform. Our aim is to understand how volunteers experience relatedness in general, and also specifically from the lens of gratitude during their work on an online volunteering platform. Understanding this can help us identify design strategies to improve volunteer wellbeing as they

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engage in unpaid work. Our study also contributes methodological on the use of codesign workshops to study volunteer experiences. In the following section, we discuss HCI perspectives on volunteer work and the role of relatedness and gratitude in volunteering platforms. We then present a case study on OSPIA, an online volunteering platform, where we investigate volunteers' perceptions of relatedness. The thematic findings suggest volunteers' perceptions for fulfilling their relatedness needs, highlighting the interplay between gratitude and other concepts. Our discussion reflects on how these opportunities should be addressed in platforms that support volunteer work.

1.1 Volunteerism and work in HCI

Volunteer work is similar to other forms of work in many ways. For instance, both volunteers and paid workers share concerns about work quality [11][12], scheduling [13][14], and the larger impact of their work [15][16]. The difference between the two is primarily the monetary gain from paid work, and consequently the underlying motivational factors and values associated with the work for volunteering [17]. Because money or other material gains do not motivate volunteers, it is important to explore motivational factors and values that HCI systems can embed to maintain good volunteer work experiences [18]. Using design methods and techniques can be useful for researching such experiences by enabling the volunteers to reflect on their perceived values [19].

It is also relevant to consider social factors for supporting volunteering work in HCI. The critical theory approach to volunteerism suggests that there is a divide between the experiences and impact of volunteerism in the Global North versus South [20]. Online volunteerism may lend itself to this divide by reinforcing existing social inequalities between the less and more technologically-equipped and technologically-abled populations [13] [14]. The shift towards issues of social justice and awareness of disenfranchisement of marginalised groups of people via technology [23] [24] is therefore, relevant, for designing for volunteerism, especially if the volunteer work aims to fairly serve all beneficiaries and be inclusive of volunteers from diverse socio-economic and demographic backgrounds and varying abilities [25].

These related works suggest that volunteering experience is shaped by several psychological and social factors that extend beyond their immediate volunteer work tasks. When these factors are addressed in designed systems via a 'volunteer-centric design' approach, they contribute to happiness and flourishing in volunteer work [3] [19]. In our study, we aim to align the concept of volunteer-centric design with that of the worker-centric design approach in HCI.

1.2 Relatedness in volunteer work

There is a history of designing for happiness and wellbeing in the HCI community. Taking a eudaimonic approach to wellbeing, engaging in enriching activities that are intrinsically pleasing to individuals in the right set of conditions leads to a life of meaning, self-realisation and genuine happiness [26] [27]. Various predictors are linked to creating those right set of conditions [28]. Relatedness, in particular, is linked with several wellbeing markers that benefit volunteers on an individual and community level [29] [30] [31].

The concept of relatedness is explored in a number of theories on belongingness [32], social connection [33], amongst others. Within HCI, Morse et al. developed *doGooders*, a social network for volunteers to engage potential volunteers and motivate existing ones and thereby improving volunteer recruitment and budget outcomes for non-profits [34]. In a study on the role of mobile social software for volunteering, Kane and Klasnja highlighted the importance of leveraging social connections and encouraging community expressions [35]. Furthermore, Farzan et al found that implementing a socialization regimen that built 'a sense of community' among volunteers improved volunteer contributions in an online community [36]. While previous studies have emphasised the importance of relatedness for online volunteering communities, they mainly focus on improving certain outcomes like increasing the number of volunteering contributions and improving volunteering opportunities. In our study, our goal is to understand how volunteers perceive relatedness with respect to volunteer work and the possible design strategies to achieve that.

1.2.1 Gratitude to improve relatedness. The field of positive psychology postulates that experiences such as compassion, empathy, forgiveness and gratitude, can enrich a person's life and enhance subjective wellbeing [37]. Gratitude involves appreciating and acknowledging the value that we receive from others [38]. Gratitude has been used in designed interactive systems to improve wellbeing and health metrics such as mood [39], body image [40], amongst others. Practicing and receiving gratitude is particularly known to mediate relatedness [38]. We found this to be true in our previous study in an online platform called *OSPIA* [3], where expressions of gratitude made online volunteers feel related to the volunteer work organisers, members of the volunteers' community, and the individuals who directly benefit from the volunteers' work (i.e. the beneficiaries) [19]. Kim et al. similarly found gratitude to be important for motivating members of an online Q&A community platform called *EnishiSource*, a system where beneficiaries who asked questions expressed their gratitude to volunteers who answered those questions [41]. In their investigation, Kim and colleagues studied gratitude as a discrete quality and found that by itself, gratitude did not significantly impact volunteer contributions. However, they noted that the feature supporting gratitude in their platform operated on a sense of obligation in the beneficiary to respond to the deed of the volunteer, thus positively impacting social interactions. Our research goes a step further, to understand volunteer relatedness from the lens of gratitude rather than studying gratitude in itself.

In this paper, we present a case study on volunteer work using a qualitative investigation. We engage volunteers in conversations about their perceptions of relatedness in general, and also via reflections on instances of gratitude. In doing so, we aim to identify opportunities to cultivate relatedness in future volunteer work.

1.3 Case study – Online Simulated Patient Interaction and Assessment Platform (OSPIA)

OSPIA is an online medical education platform used to train and assess medical students' communication skills and interactions with

patients at an Australian university. In this teleconferencing web platform, volunteers play the essential role of simulated patients (SPs) in mock interviews with the medical students. Volunteers use OSPIA for registration, training, appointment booking, student-SP interview, and student assessment. The SPs are provided with a script which includes medical, social and personal details of the patient that they are supposed to characterize in their mock interviews. A similar face-to-face module is held on campus and runs parallel to the OSPIA module. The online and the face-to-face modules are similar in most ways, however the information and communication needs are completely integrated into the OSPIA platform for online volunteers while for face-to-face volunteers, a combination of phone, email and postal service are used.

2 METHODOLOGY

We conducted four co-design workshops with a total of nine volunteer SPs from both OSPIA (four SPs) and the face-to-face modules (five SPs) – two workshops with face-to-face SPs and the other two with the OSPIA SPs. We included both groups because this was found beneficial in our previous studies in order to generate deep insights [3] [19]. Workshops were moderated by an HCI researcher. All participants provided written informed consent. The study protocol was approved by an ethics committee (ref HC16048).

Using generative tools and craft material, we developed activity worksheets around the SP experiences of relatedness and gratitude. The worksheets captured the timeline and key steps in a typical student–SP interview. The SPs reflected on their overall work and not just the student-SP interview session so as to capture their overall experience. For each workshop, the participants were asked to recall and identify on worksheets a memorable experience from their time as an SP where they (i) felt connected with the student (relatedness activity), and (ii) felt appreciated by the student (gratitude activity). A set of stickers was provided to participants to capture their affective experiences associated with those moments. These stickers consisted of popular emoticons (e.g. happy/sad face) as well as graphical expressions of feelings in our previous study (e.g. helping, appreciation) [3]. The generative tools were designed to aid subsequent participant discussions on the topics rather than being a validated tool for assessment (see Figure 1).

Next, the participants were engaged in a facilitated discussion to expand on the key moments captured in the worksheets. These discussions were audio recorded, transcribed and analysed thematically, using interpretative phenomenological analysis [42]. Through a bottom-up analysis, we generated codes, and meaning-making was guided by interpreting the underlying themes. For instance, participants' satisfaction arising from the student's desirable formal behaviour was coded 'etiquettes', and participants' appreciation of others being conscious of their time and efforts was coded 'consideration'. These were later combined to form the theme 'courtesy'. Likewise, all the text under the codes 'appreciation' and 'encouragement' was categorised under the theme 'reciprocation'. All worksheet activities were combined for analysis as we aimed to understand relatedness on its own and via instances of gratitude.



Figure 1: Volunteers in the workshop (top), worksheets used in co-design workshops (down)

3 RESULTS

The analysis yielded seven themes highlighting volunteers' perceptions and values in relation to their relatedness experiences, as well as the corresponding opportunities, as discussed below.

3.1 Courtesy

This refers to the volunteers' notions of deservedness of respect for their volunteering efforts by the beneficiaries (i.e. students) and the program organisers. Participants identified two opportunities for expressing courtesy. Firstly, participants discussed their preference for general formality in interactions with students, such as presentation, professionalism, and preparedness shown towards the volunteers. "The reason I picked this one [memorable experience] was because the student was generally well prepared. She was very interested. . . She was really impressive. She seemed calm. It was her presentation. She listened and she was organised." OP1.

The second opportunity was linked to valuing volunteers time. "It [the appointment booking process, which allows booking 72 hours at the most in advance to the SPs] could be made a little bit better. . . Maybe I could offer a little bit more [appointments], but having to do it 72 hours, I have to be really sure that afternoon I am going to keep it open." OP1. "I don't volunteer to get the kudos. But I think if the student just acknowledges that you have given me the time, that is really quite nice... It is part of politeness, manners, that sort of thing." OP2

3.2 Pride in their work

Our participants discussed the importance of being able to do everything in their power to help the students, the beneficiaries of their volunteer work. The impact that their volunteering work has on the students is a source of pride for them. "The nice thing is that I will often see first year [students] and they come back their second year . . . and I will often say, 'I remember you, my God you have come a long way!' It's fabulous!" CP3. In line with taking pride in their efforts, the SP volunteers strive to be prepared and presentable for their volunteering work. "In terms of the actual student assessment, I felt very pleased that I had prepared, I had been given the role a week in advance, I had prepared because it

wasn't entirely easy." CP1. Participants also noted the opportunity for the OSPIA system to help indicate this impact via an immediate student feedback on their volunteer contribution to help build their relatedness, as demonstrated by the following conversation:

OP1: "I guess there is an opportunity there – 'Thanks, I have noted your comments' – Maybe in that particular it comes back to us. What do you think?". OP2: "That would be great! I would love to see what they thought of how the interview went. They rate us in a similar fashion". OP1: "Yeah, and even if they commented on things that they felt hadn't gone so well. Like just a little bit of back and forth".

3.3 Reciprocation

When their volunteer work was reciprocated via explicit acknowledgement and feedback by the students or the program organisers, the volunteers felt appreciated. Our participants indicated the opportunity that reciprocation can be designed and be directed at them as individuals. "It's the same student [referring to online acknowledgement], the fact that she thanked me in the end, I felt valued." OP1.

They also indicated that reciprocation can be directed towards volunteers as group (which they self-identified with). "We got an invitation to go to the university with the other OSPIA simulated patients as a thank you and I thought that brought it home to me cause we actually got to meet Silas and Kiran [the organisers]." OP2.

3.4 Managing expectations

Our participants identified that providing adequate information at different points of their volunteering journey helps set expectations which can then provide seamless and relatable experiences for them. For instance, adequate onboarding augmented with information at the beginning of the program is important in building their initial expectation about the program, e.g. while referring to the lack of clarity in the initial description of the program, an SP mentioned: "I was glad that I was accepted, because the information that I saw wasn't very descriptive." OP2

Additionally, the upkeep of regular communication regarding the OSPIA community may help retain volunteer interest in the program and build relatedness.

"We also had the opportunity to find out a little but more about the program. But we didn't really know, like facial recognition and some of the other things. I had no idea about the enormity of the program. I knew that obviously a lot of students doing undergrad for medicine at UNSW, but I did not know how many of us there were". OP2

They noted that knowing what is expected from the volunteers would reduce ambiguity in their relationships "So, it is a two-way street. You can talk about how we get the acknowledgement weekly, but actually getting to meet people and know where they are coming from and also what their expectations are of us too. That meant a lot to me." OP2

3.5 Reflecting together

A number of participants expressed pleasure in sharing thoughts and reflections with students and vice-versa. They identified the value of enabling personal growth through this. For instance, opportunities for reflections as part of work may allow the volunteer to understand the student's perspectives. This may cultivate growth for both the volunteer and the student. "Maybe they could have the option to reflect, on what you have told them, see if they thought if it was useful or not, or were they worried about other things that went on." OP1

Participants also mentioned how the casual nature of such reflections led to rapport building via sharing personal stories, informal banter and small talk. "If we had time at the end, they [student] would ask, 'What do you usually do?' They were interested in us as well! CP2".

3.6 Taking charge of their volunteering work

Our participants noted that taking control of their volunteering tasks and personalising their performance helps them express themselves and their individuality which then reflects on their connection with others. Giving the volunteers flexibility in the means and methods of performing their volunteering task was preferred. "What I find a bit more difficult, I guess more if it's not a straightforward scenario, is that you don't get a lot of time to memorise. You don't get a lot of time to remember it and I am not great at learning things quickly. I just try and remember the name bits and just freeball after that. Like come up with my own stuff." CP5

3.7 Community building

Participants reflected on the importance of access to a supportive community that they identify with. Building belongingness to a wider group of volunteers can help volunteers gain peer-support and build confidence. "It would be really nice to talk to like-minded people. I thought it would be nice to talk to socially, but also find out where they are coming from. When you take on something like this, you know you are on your own, but I guess attending one of those things [an organisational event to socialise volunteers] you realise you are reinforcing the issues that you are having is same as the other people." OP2

Additionally, integrating non-task related interactions with other volunteers, students, and organisers contributes to community building. "Cause I guess if you wanted to use an analogy, in this particular it is recruiting the cart before the horse. Because we are doing the program and then we are meeting the people [via organisational events]." OP2.

In the following section, we discuss the results in light of certain matters that elucidate our previous argument about social and psychological factors (section 1.1) that are relevant to volunteering in OSPIA, and also to provide further insights for building relatedness in other online platforms that use volunteer labour.

4 DISCUSSION AND CONCLUSION

4.1 Formal v/s informal relations

Delving deeper into the nuances of volunteer experiences that were evident in the themes, we find a contrast between the volunteers'

desire for formal vs informal connections with the beneficiaries and others within the community. Growth and flourishing were cultivated in volunteer experiences through sharing reflections when they had the opportunity to get to know others, empathise with them, and share personal stories, rather than formal work tasks. However, they recognised the value of formalities and wanted professionalism, respect and reciprocity for performing the volunteering work. We propose that future design strategies should foster both, however we acknowledge that achieving a balance between the two may be challenging. Achieving that kind of balance takes a holistic perspective that designing for social interactions in volunteering platforms should aim for.

4.2 Emotional labour in volunteering

Our study on volunteer perceptions and values in relation to relatedness points towards emotional investment of the volunteers in their work, which manifested itself in themes such as courtesy, pride in their work, reciprocity and others. As the volunteers in our study pointed out, this emotional labour must be recognised by the beneficiaries as well as others involved in the program. Emotional labour was termed by Hochschild who established that some forms of work demand more emotional efforts than others, e.g. customer facing workers, flight attendants [43]. Raval and Dourish discussed affective and emotional aspects of labour in online crowdsourcing platforms that use volunteers [44]. Online volunteering platforms, especially the ones where the volunteers perform emotionally demanding work, e.g. disaster and crisis response [45], counselling [46], moderating online communities [47], do not always adequately address the emotional labour of these volunteers. This is an underexplored opportunity which we plan to address in future research.

Moreover, our previous research has highlighted the higher participation of female volunteers in the OSPIA program and its face-to-face counterpart [3]. Related research has shown that in most instances of face-to-face volunteering, women usually outnumber men, rank higher in prosocial traits and are usually more motivated (than men) to provide help [48]. However, this trend doesn't always follow in all online spaces. Wikipedia, for instance, is known for its lack of gender diversity among its volunteers, with women forming less than 15% of Wikipedia volunteers. Assumptions about lack of technical skills in women were previously shown to be responsible for this trend [49]. However, a study by Menking & Erickson attributed it to gendered emotional labour in Wikipedia, as women termed 'Wikistress' to describe the emotionally volatile and unsupportive atmosphere which drove them to quit volunteering on the site [50].

We believe that future strategies in designing online volunteering platforms should recognise the emotional labour of volunteers (particularly those who belong to under-represented groups, such as women) in addition to the cognitive and other aspects of labour. The outcome will be online volunteering inclusive of not only women, but other genders and groups who are emotionally committed in online volunteering work as well.

Our findings have implications for improving relatedness experiences in online civic engagement initiatives, of which online volunteering plays an important role [51]. The importance of

designing better relatedness experiences for volunteering work in HCI is underlined now more than ever, with the COVID-19 pandemic sweeping the globe into physical distancing measures, thus increasing the reliance on virtual technology for interpersonal and community participation ventures [52].

Our findings have clear implications for design of online volunteering platforms and platforms with voluntary participation of people where there is a component of volunteer-beneficiary interaction such as OSPIA. However, most findings can be generalised to platforms with no or subtle indication of volunteer-beneficiary interaction such as the United Nation's online volunteering website [53] where building relatedness could focus on the organisers i.e. United Nations and its relevant partners who benefit from the volunteers' work.

5 FUTURE WORK

Future work will include developing prototypes of the OSPIA platform based on the opportunities and strategies presented in this paper. In particular, we will explore emotional labour as a mandate for improving online volunteering experience, specifically for identified marginalised groups.

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